Lab 1: Transient Heat Conduction in a Fin

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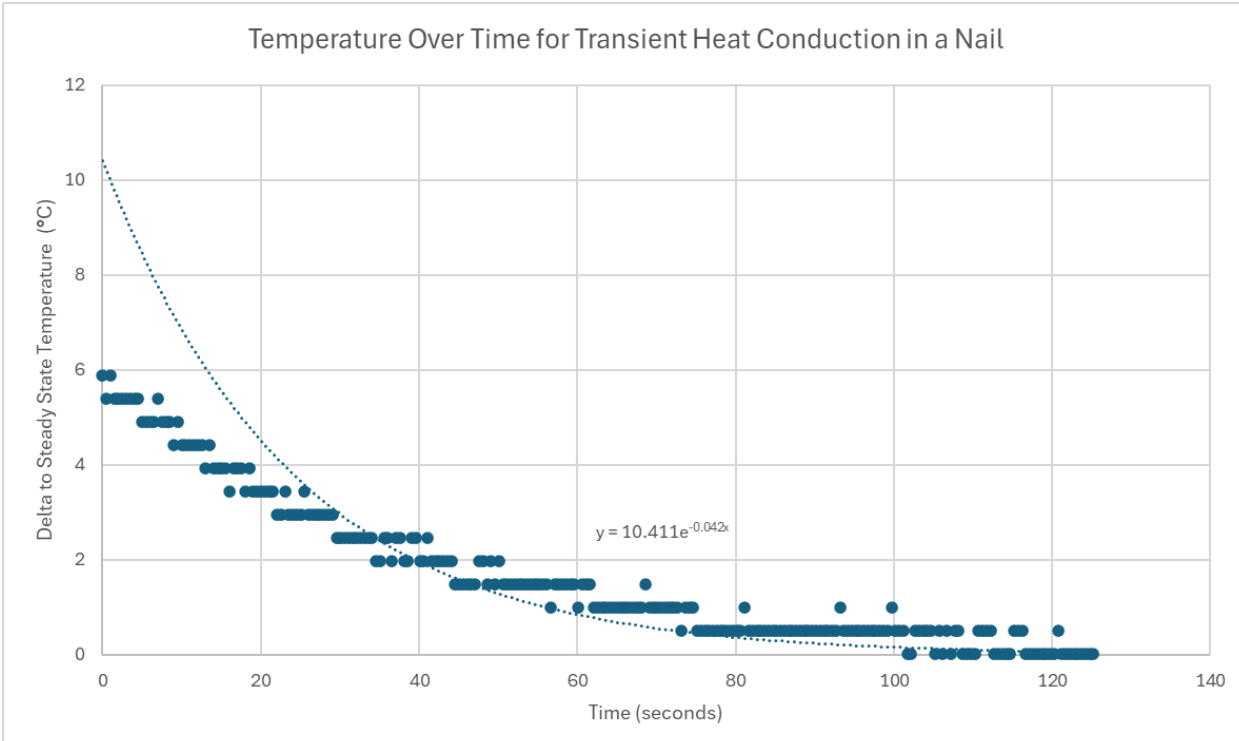
This lab aims to simulate transient heat conduction in a fin by immersing the bottom of a nail in ice water and recording the changing temperature at the top of the nail. A circuit with Arduino and a temperature sensor was built to record the temperature, and an Arduino script was written to collect data. The top of the nail was superglued to the temperature sensor, while the nail tip was immersed in ice water for the experiment setup. The Arduino script is shown below:

```
const int kPinTemp = A0;
unsigned long time;

void setup() {
  Serial.begin(9600);
}

void loop() {
  float tempC = getTemperatureC();
  time = millis();
  Serial.print(time);
  Serial.print(" s:");
  Serial.print(tempC);
  Serial.println(" degrees C");
  delay(1000);
}

float getTemperatureC() {
  int reading = analogRead(kPinTemp);
  float voltage = reading * 5.0 / 1023;
  float tempC = (voltage - 0.5) * 100;
  return tempC;
}
```

The Arduino function “millis()” was used to record data as a function of time and serial monitor was used to read data from the temperature sensor. The function getTemperatureC() was written to convert the raw analog signals from the temperature sensor into a temperature value in Celsius. The Arduino reads the sensor output voltage as an integer between 0 and 1023, which corresponds to an input voltage range of 0 to 5V. Therefore, the reading was multiplied by 5 and divided by 1023. Subtracting 0.5 removes the voltage offset as the sensor outputs 0.5V at 0 degrees Celsius.

The results are shown in the scatter plot below. Our steady state temperature was found to be 15.49°C. An exponential trendline was fitted to the plot and its equation is $y = 10.411e^{-0.042x}$.



We want to find the convective heat transfer coefficient and the thermal diffusivity for our experiment. Given that our room temperature $T_\infty = 21.36$°C, steady state temperature $T(L) = 15.49$°C, and ice water temperature $T_0 = 0$°C, we can use the equation

$$T(L) = (T_0 - T_\infty)\frac{1}{cosh(mL)} + T_\infty$$

to solve for m and use $m^2 = \frac{\bar{h}P}{KA}$ to solve for the convective heat transfer coefficient $\bar{h}$, where L is the length of the nail (3.9 cm), P is the parameter of the nail (0.002π m), and A is the area of the nail top ($0.001^2\pi\, m^2$).

$$15.49 = (0 - 21.36) \cdot \frac{1}{cosh(m \cdot 0.039)} + 21.36$$

$$m = 50.46$$

$$(50.46)^2 = \frac{\bar{h} \cdot 0.002\pi}{47 \cdot 0.001^2\pi},\ \bar{h} = 59.84\ W/m^2K$$

Our value of $\bar{h}$ is consistent with the recognized convective heat transfer coefficient for steel surfaces in contact with liquids, which ranges from 50 to 10,000 W/m^2K.

To solve for thermal diffusivity α, we quote the solution to question 2 of Problem set 6, which shows the analytical solution for transient heat conduction in a straight (1-D) fin, with steady state and transient eigenfunction expansion. The equation shows the following:

$$T(x,t) = T_\infty - \frac{T_\infty \cosh(m(L-x))}{\cosh(mL)} + \frac{2T_\infty}{L}\sum_{n=1}^{\infty} \frac{\lambda_n}{m^2+\lambda_n^2}\sin(\lambda_n x)e^{-\alpha(m^2+\lambda_n^2)t}$$

where $T_\infty - \frac{T_\infty \cosh(m(L-x))}{\cosh(mL)}$ is in a steady state. If we assume that the dominant term of the Riemann sum is the first term, then the measured temperature difference can be approximated as follows:

$$y(t) \approx Ce^{-\alpha(m^2+\lambda_1^2)t}$$

Matching this to the equation from our graph, we can see that

$$0.042 = \alpha(m^2+\lambda_1^2)$$

$$\alpha = \frac{0.042}{m^2+\lambda_1^2}$$

We would need to fit our solution to the boundary condition to solve for λ. Since we are not given a specific boundary condition, we would assume an insulated tip, where,

$$\lambda_n = \frac{(2n-1)\pi}{2L} \Rightarrow \lambda_1 = \frac{\pi}{2L} = \frac{\pi}{0.078}$$

Thus, we can then approximate α:

$$\alpha = \frac{0.042}{50.46^2 + (\frac{\pi}{0.078})^2} = 1.008 \cdot 10^{-5} m^2/s$$

Published values for the thermal diffusivity of steel at room temperature range from roughly $4.2 \cdot 10^{-6} m^2/s$ (for stainless steels) up to $1.88 \cdot 10^{-5} m^2/s$ for 0.1% carbon steels [1]. The experimental thermal diffusivity is inside the range, which suggests our results are reliable.

Sources:

[1]: https://www.engineersedge.com/heat_transfer/thermal_diffusivity_table_13953.htm